

Observation



wrist RGB

s^p

ℓ : pick up the kettle

house0, same timestep

Frozen VLA + RL token

MolmoAct2 (frozen)
VLM + flow action expert

last-layer tokens $\mathbf{z}_{1:M}$



g_ϕ (then freeze)

$z_{rl} \in \mathbb{R}^{256}$

$\tilde{a}_{1:C} \sim \hat{\pi}$

$$z_{rl} = g_\phi([\mathbf{z}_{1:M}, e_{rl}])_{M+1}$$

solid orange: tokens dashed green: $\hat{\pi}$

CFGRL extractor

RL state $x = (z_{rl}, s^p)$

critic Q_ψ
ensemble, chunk TD

flow v_θ
 $v_\theta(x, a_t, t, o)$
 $o \in \{\emptyset, 1\}$

$$A(x, a) = Q_\psi(x, a) - Q_\psi(x, \tilde{a})$$

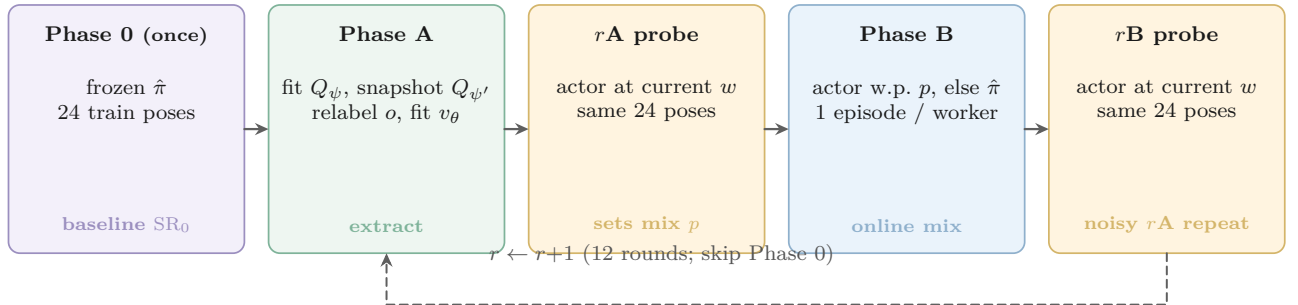
$o=1$ if $A \geq 0$ or success; else $\emptyset$

$$v = (1 - w) v_\theta(\cdot, \emptyset) + w v_\theta(\cdot, o=1)$$

$$\pi \propto \hat{\pi} p(o=1 \mid x, a)^w$$

$w=0$ then 0.5

Iterated extraction (CPI)



Critic (always on)

$$\mathcal{L}_Q = \mathbb{E}[(\hat{Q} - Q_\psi)^2] + \lambda_{CQL} \mathcal{L}_{CQL} + \lambda_{rank} \mathcal{L}_{rank}$$

$$\hat{Q} = r_{1:C} + \gamma^C Q_{\psi'}(x', a')$$

o from stop-grad $Q_{\psi'}$

Actor (Phase A only)

$$\mathcal{L}_{CFGRL} = \mathbb{E}[\|v_\theta(x, a_t, t, o) - (a^* - a_0)\|_2^2]$$

$$a^* = \tilde{a} \text{ if } o = \emptyset$$

$$a^* = a \text{ if } o = 1$$

10% dropout to $\emptyset$

Deploy w and mix p

$$w = 0 \text{ until } \|v_\theta(\cdot, \emptyset) - \tilde{a}\| \leq 0.02$$

then $w = 0.5$

$$p = 0.25 \text{ if } SR_{rA} \geq SR_0 - 1/n$$

else $p = 0$ (execute $\hat{\pi}$)

$n = \#$ probe episodes; p uses rA only